

# Degly Sebastián Pava Pava

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<https://github.com/DglyP>

Systems and Computer Engineering

## WORK EXPERIENCE

From: January 2018  
To: Present

### HIVE Research Lab (Dr. Claudia Krebs) – ELAP - University of British Columbia

Worked on-site and currently working remotely in neuroanatomy-based applications while making use of Mixed Reality technologies for educational and research purposes. A Holographic brain is being used to help students understand its parts and how it works by using the Unity Engine.

- Part of Holobrain 2.0 development team
- Volume Rendering for MRI images in the Unity Engine using HoloLens 1
- Emerging Leaders of the Americas Program Scholarship winner, offered by the Canadian Government

From: February 2017  
To: September 2019

### Simulation and Modelling developer at Pontifical Xavierian University

- Department of Industrial Engineering and Systems Engineering at Pontifical Xavierian University, Bogota, Colombia.
- By using the NetLogo programming language simulate a production planning, scheduling, and inventory control system used to manage manufacturing processes, also known as a Material Requirements Planning. It is currently being used in Production Classes to teach new students about MRP and Scheduling. Currently remote consulting for the projects.

From: November 2015  
To: January 2016

### Designer/Coder and publisher for ASSET-TUTORES Website [www.asset-tutores.com](http://www.asset-tutores.com)

- Created and designed an intuitive website to promote the company and communicate all its services. Currently published at [www.asset-tutores.com](http://www.asset-tutores.com).

## EDUCATION AND TRAINING

### Research Groups:

*Grupo de investigación en procesos Químicos y Bioquímicos (Rubén Darío Godoy Silva) – Universidad Nacional de Colombia*

From: February 2016  
To: December 2017

Chemical and Biochemical Processes research group

- Programmable Logic Controllers to automate a machine or process

From: June 2017  
To: December 2018

### RCODI's Research Group (Professor Sabine Brunswicker) – Purdue University

- Lead developer of their Energy Modeling Tool based on AWS, HTML and Matlab Cloud Computing while also handling the Purdue-Unal IronHacks 2017/2018 by teaching and helping with the contest.

### Training:

From: January 2016  
To: Present

### Teacher for school and college students with ASSET-TUTORES

- Give out classes for school and college students in need to improve their knowledge and their skills.
- Taught English, Mathematics, Website Building and other Computer and Software related subjects

### Bachelor studies:

*Special Admission Program, best high school graduates of Colombia Scholarship at Universidad Nacional de Colombia.*

## PERSONAL SKILLS

Spanish  
English  
Korean

Native  
Advanced - TOEFL iBT web-based test (C1 – 103 points)  
Basic Writing Skills

### Technical skills

#### Platforms and Programming Languages Experience:

Windows, Linux, Unix

|            |         |        |          |          |          |
|------------|---------|--------|----------|----------|----------|
| NetLogo    | 7 years | Java   | 4 years  | Angular2 | 1 year   |
| HTML       | 6 years | Python | 4 years  | Ruby     | 1 year   |
| JavaScript | 4 years | React  | 2 months | C#       | 6 months |

## AWARDS AND PROJECTS

### Android and iOS app developed in Flutter

Currently developing a mobile application focused on improving traffic and quality of life in Bogotá. Based on UX and H-CI principles.

### Energy Modeling Tool – Purdue's Research Center for Open Digital Innovation

HTML and AWS based software focused on helping civil engineers and architects find the best way to build an energy efficient building by using data obtained from EnergyPlus processing, Open Data and Matlab Cloud Computing for costs calculation.

### Winner of Purdue-UNAL Hackathon - Gold IronHacks 2017

Developed a website with a mashup that utilized open data to help students rent houses near the University of Illinois, Chicago. Open Data and D3.js were used to create interactive visualizations to give insights and support optimal decisions. The project was developed through Git and judged by a team of experts from Purdue University, including Dr. Sabine Brunswicker.

### Lead developer, Simulation of TransMilenio

Developed a NetLogo simulation of the Bogotá's transport service Transmilenio to try to optimize how buses should be deployed to improve the quality of service. The software simulated how the system works and proposes an algorithm to improve its quality.